Blockchain Dapp Proposal for “Fikra” Competition:

Ministry of Interior (Traffic Directorate)

SijilChain

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Group 07 - Sec 01

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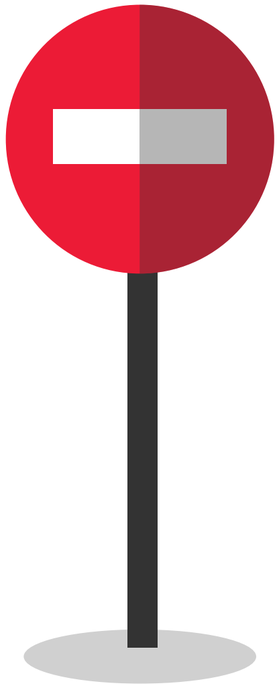

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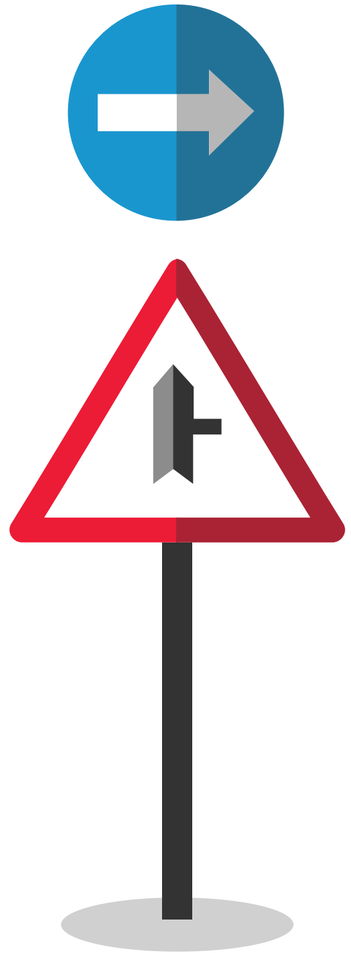
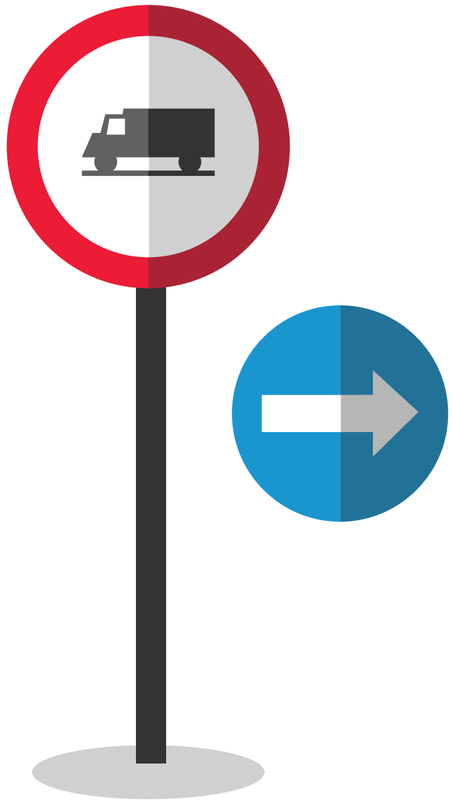
Introduction

SijjilChain is an app built on **blockchain** for the use of the **Ministry of Interior.**

It’s a system which provides **improved transparency & trust** in the second-hand car market by giving **users access** to a full & validated **car history** via the VIN / Chassis No.

Blockchain is utilized as a method to **store vehicle records securely**, forever linking them together and making it **impossible** to alter/hide them.





02 Problem Definition

Inside this section:

- **Problem Definition**
- **Main Challenges**
- **The Core Problem Concept**

Problem Definition

- **File numbers** have **priority** when referring to a vehicle's records but not the **VIN (chassis number).**

- If a vehicle is labeled as a **total loss,** then the **same vehicle may:**

  - **be restored**
  - **registered into a new file.**

-The **new file** will not have the vehicle **past accident records,** resulting in the **"clean record illusion"** problem.

Main Challenges

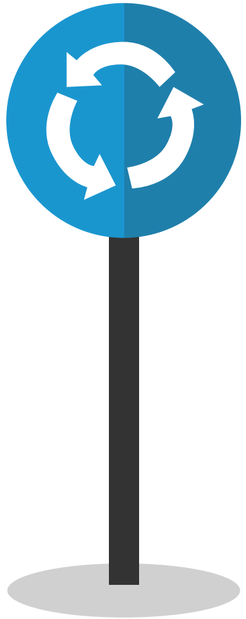
Buyers being fooled to purchase dangerous cars

Vehicle history is concealed or inaccurate

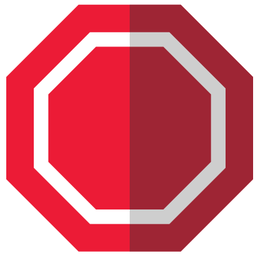
Requires manual check for verification (search via VIN at the traffic departments)

Consequences that arise due to the problem:

- Financial losses
- Safety hazard
- Market loses credibility

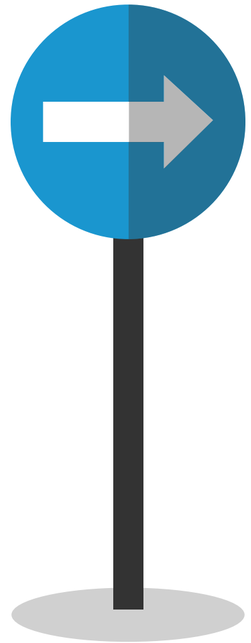


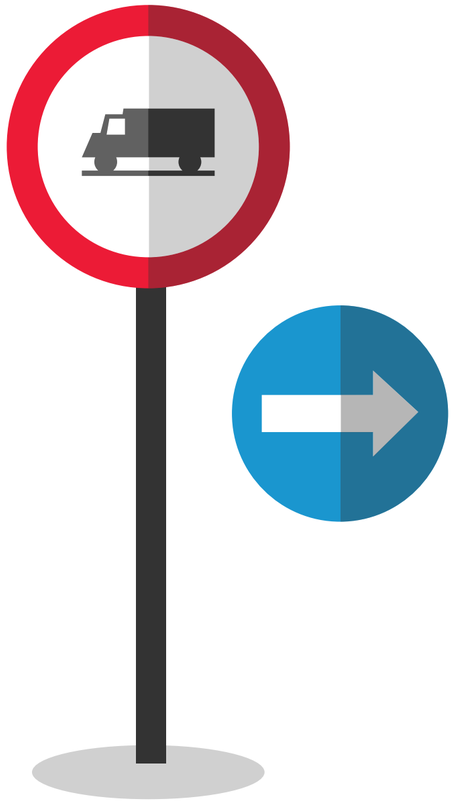
The Core Problem Concept



VIN is permanent but the files are not.

The current system supports data reset.



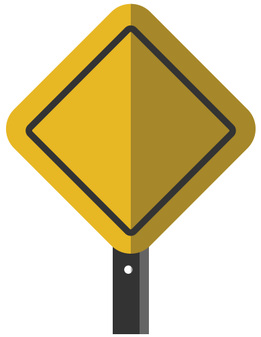


03 Context & Stakeholders

Inside this section:

- **Context**
- **Stakeholders**

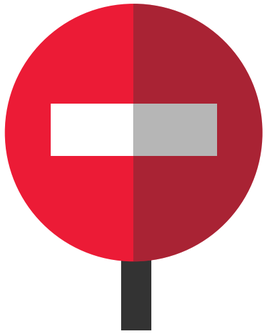
Context



The problem is found at the

Ministry of Interior (Traffic

Directorate)



Responsible for the following:

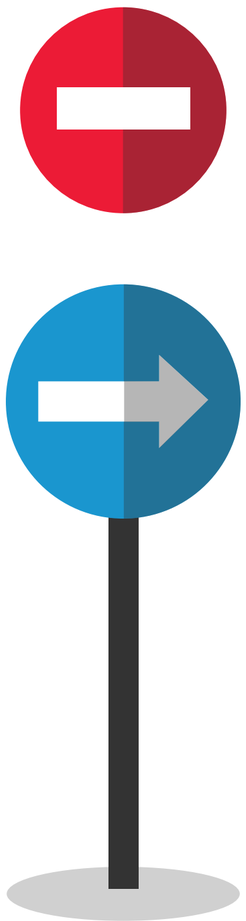
- Vehicle registration
- Accident records
- Cancellation records

The current system

has issues

with transparency.

Stakeholders



Used Car Buyers

- Need an honest representation of the car's history
- Risk being cheated by incomplete records

Car Sellers

- Need a solution to increase their credibility.

Traffic Directorate

- In charge of maintaining official records
- Needs a credible system

Insurance Companies

- They need reliable integration.
- Will provide accident and total loss information.

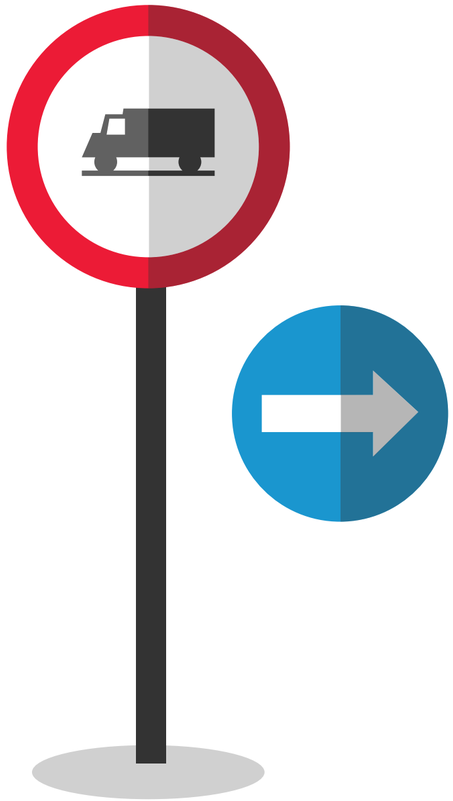
Inspection Centers

- Provide vehicle status information.
- Contribute to lifecycle of vehicle.

Government / Auditors

- In need of complete audit trial

04 Proposed Solution



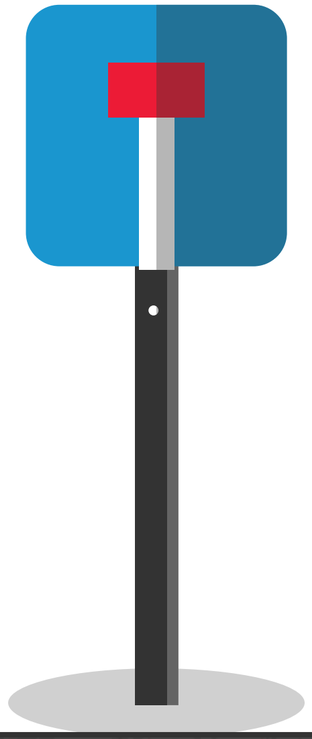
Inside this section:

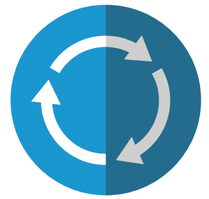
- **Solution Description & Concept**
- **System Functions & System Key Features**
- **The User Benefits from the system & The System Benefits**

Solution Description & Concept

Solution Description

- SijjilChain is a DApp based on Blockchain technology
- Is designed for vehicle history transparency





Concept

- Using the VIN number of a vehicle to connect all its records together (not files' numbers).

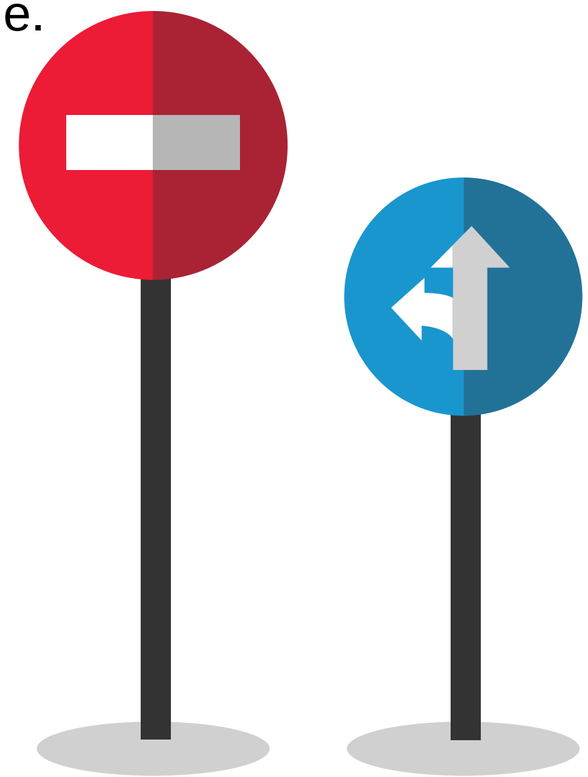
System Functions & Key Features

Connects all the necessary data using the VIN number:

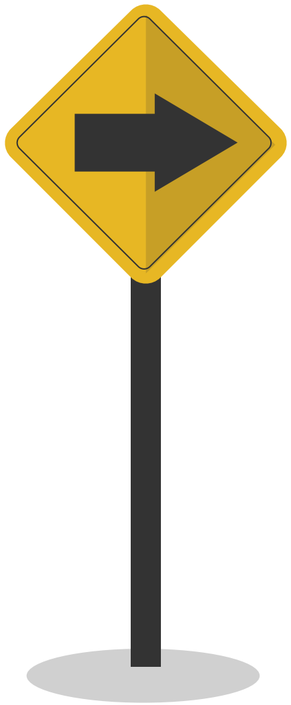
- Car accidents, registration cancellations, repairs, registration of new file.
- Creates comprehensive history of the vehicle.

System Key Features:

- Blockchain technology data management
- Irrevocable and tamperproof database
- Complete tracking of whole lifecycle of a vehicle
- Mobile application interface
- No need to visit traffic centers



The User Benefits from the system & The System Benefits

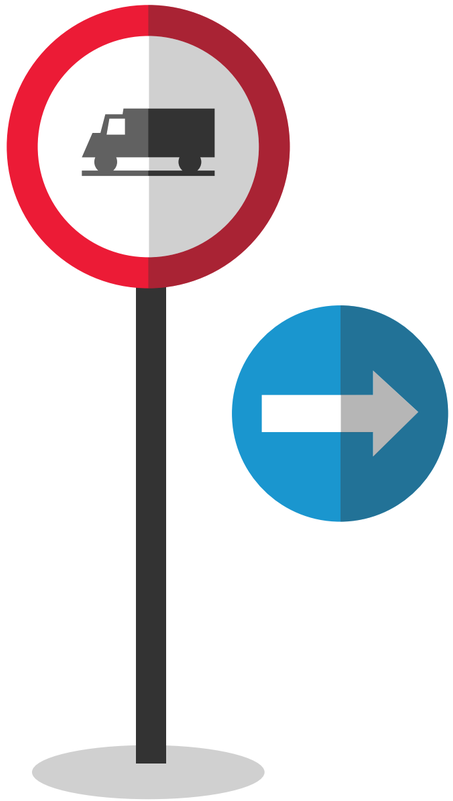


The User Benefits from the system:

- Verify vehicle history
- Detect possible fraud
- Make sound purchase decisions

The System Benefits:

- Eliminate possibility for data reset fraud
- Increase credibility of used car market



05 How the Solution Works

Inside this section:

- **The system workflow**

How the Solution Works

Step 1

Enter VIN : The buyer provides the vehicle Vehicle Identification number.

Step 2

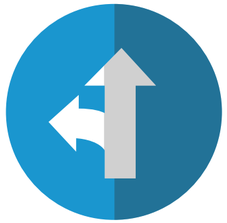
Send OTP for Authorization: The system sends OTP to seller for authorized access.

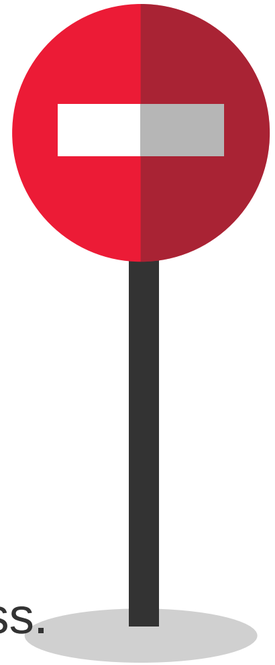
Step 3

Seller Provides OTP: Seller provides the OTP to the buyer.

Step 4

OTP Verification: 1- OTP Verification takes place on the backend side.2- Authorization confirmation process.





How the Solution Works

Step 5

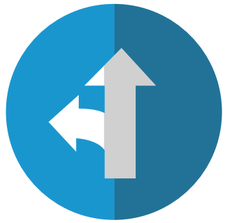
Retrieve Blockchain Records: including:
1- Accidents records 2- Cancellation records.
3- Previous files' records

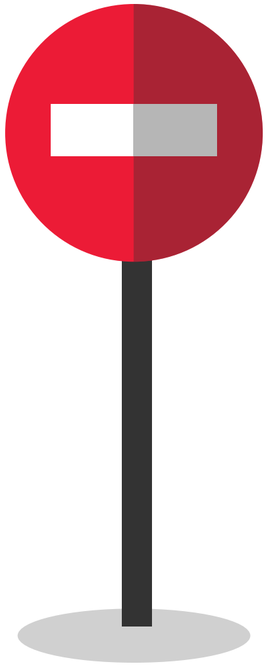
Step 6

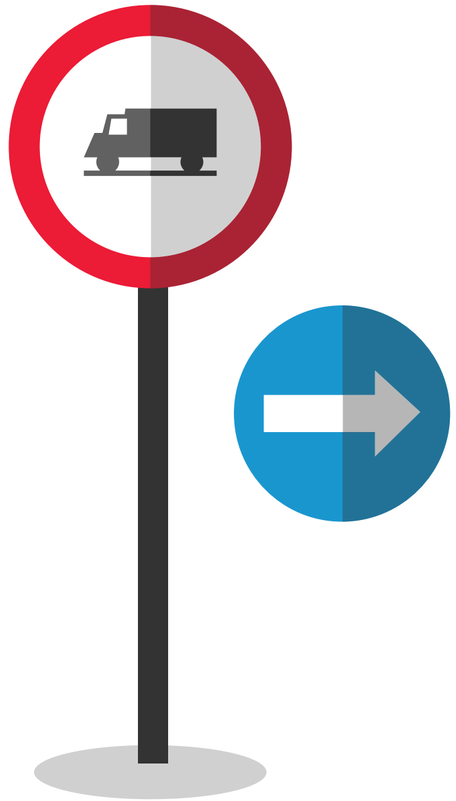
Display Results: App displays: 1- Vehicle summary 2- Accidents' details 3- Cancellation details 4- Assessment of safety risk: safe, warning, dangerous 5- Blockchain records verification

Step 7

Decision Making: Based on the verified information, the buyer: 1- Buys the car 2- Negotiates its price 3- Rejection of the car



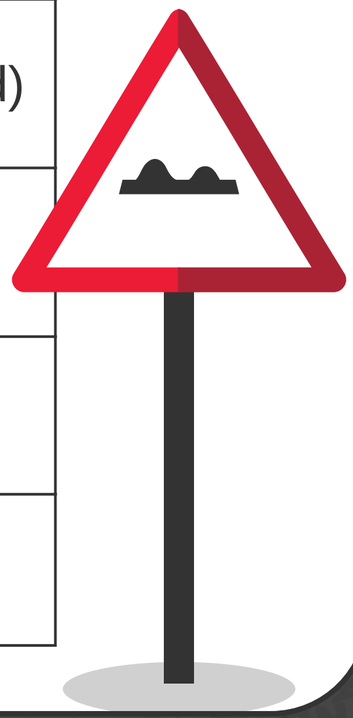


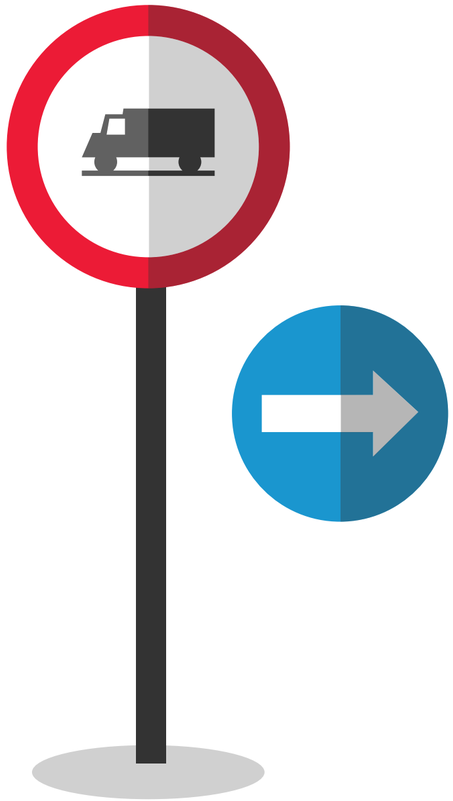


06 Blockchain Justification

Blockchain Justification

Traditional System	SijjilChain (Blockchain)
Records stored in separate files	Records linked using VIN
Data can be hidden or manipulated	Data is immutable (cannot be changed)
Limited transparency	Full transparency of vehicle history
Centralized control	Decentralized and secure
Difficult to trace past records	Complete traceability of all updates





07 Technical Framework

Inside this section:

- Overview
- Blockchain Design
- Smart Contracts
- Data Flow
- User Flow
- System Architecture Diagram

Overview

Layer 1 - Front End

- User Interface of the Mobile Application
- Input VIN to display Results

Layer 2 - Back End

- Server API (requests)
- Processes and communicates with Vehicle Verification Service

Layer 3 - Blockchain

- Records all vehicles
- Immutable and transparent

Layer 4 - Smart Contracts

- Operates on Vehicle Data
- Linking Records via VIN

Layer 5 – Optional Off-Chain Storage

- Large or Non-Critical Data
- Better Performance

Blockchain Design

Blockchain Type

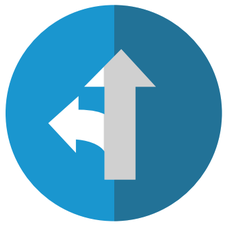
-blockchain is permissioned/ private

-controlled by public or private organizations (such as law enforcement agencies or other government entities)

Platform

-Hyperledger Fabric

-or Private Ethereum platform



Consensus Mechanism

-proof of authority (PoA)

-or Practical Byzantine Fault Tolerance (PBFT)

Data Structure

-VIN identifies each vehicle.

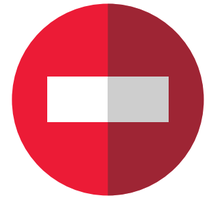
-new transactions for each data update are created and added to the blockchain.

Smart Contracts

Smart Contracts for Vehicle Data Management Using Blockchain

Primary Functions:

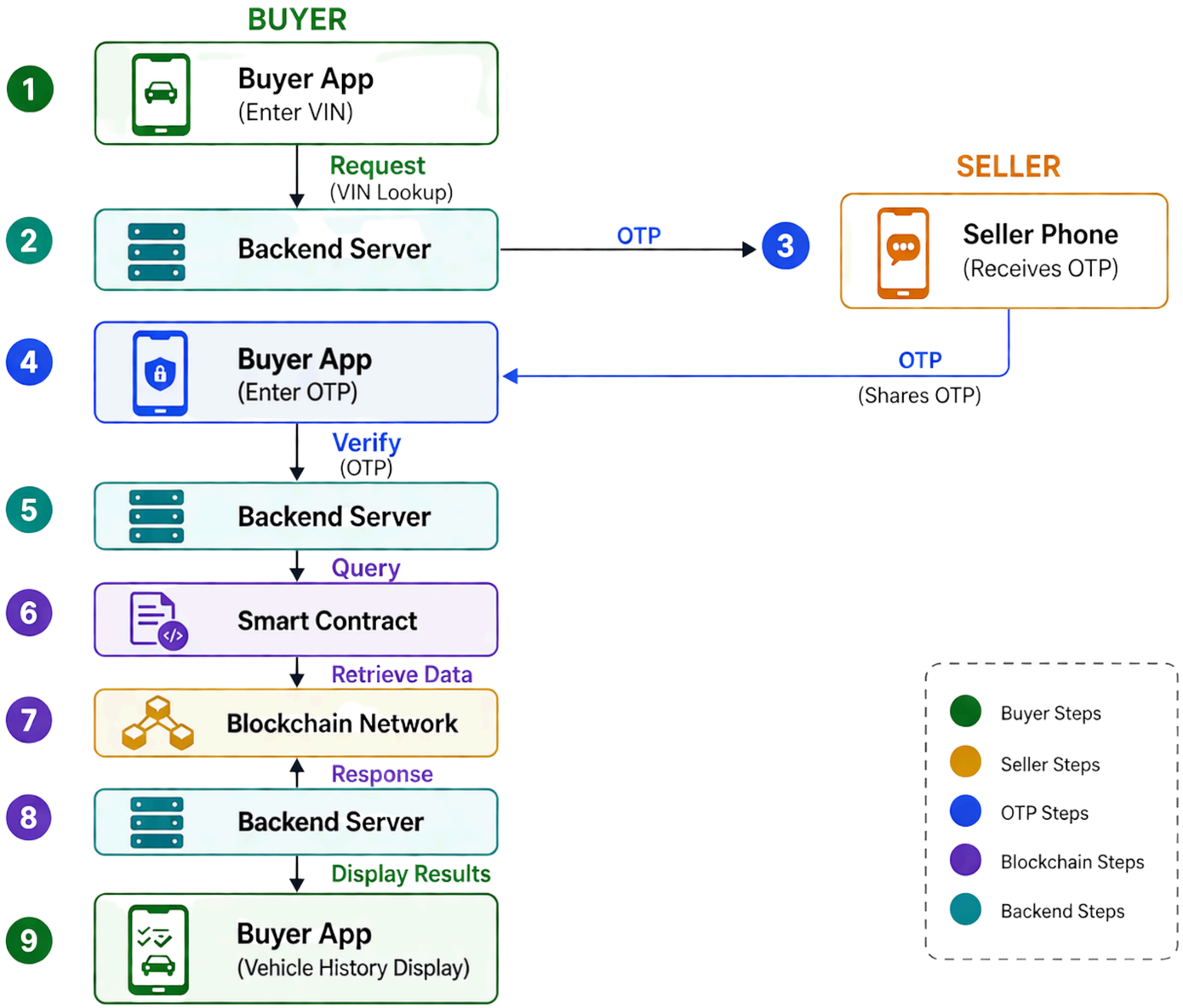
- Use a VIN to register your car
- Enter an accident record into your vehicle's history
- Cancel your vehicle
- Associate a new registration with an already registered VIN
- Retrieves all vehicle history



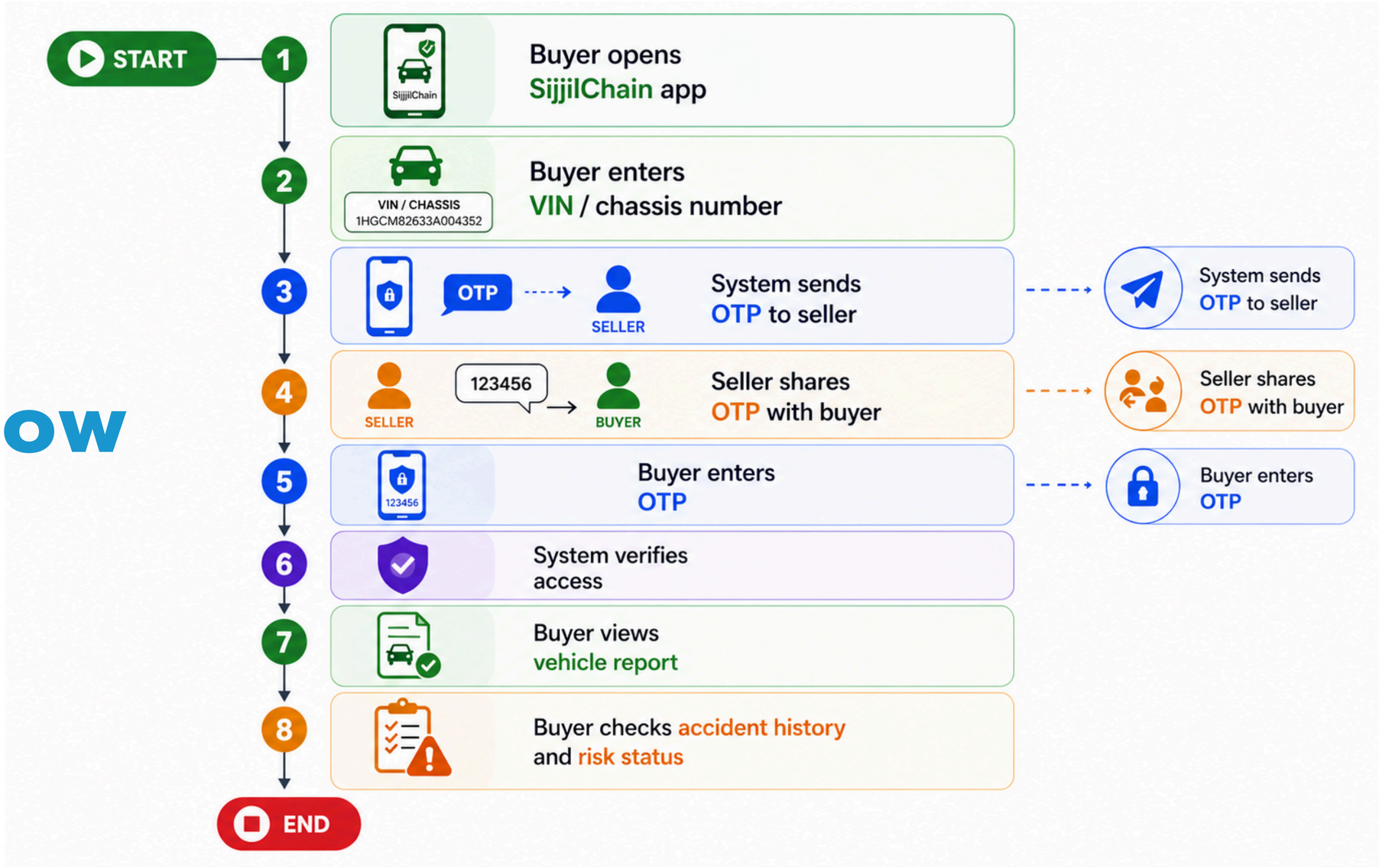
Key Features:

- Automates action in smart contract
- Is secure, and cannot be altered

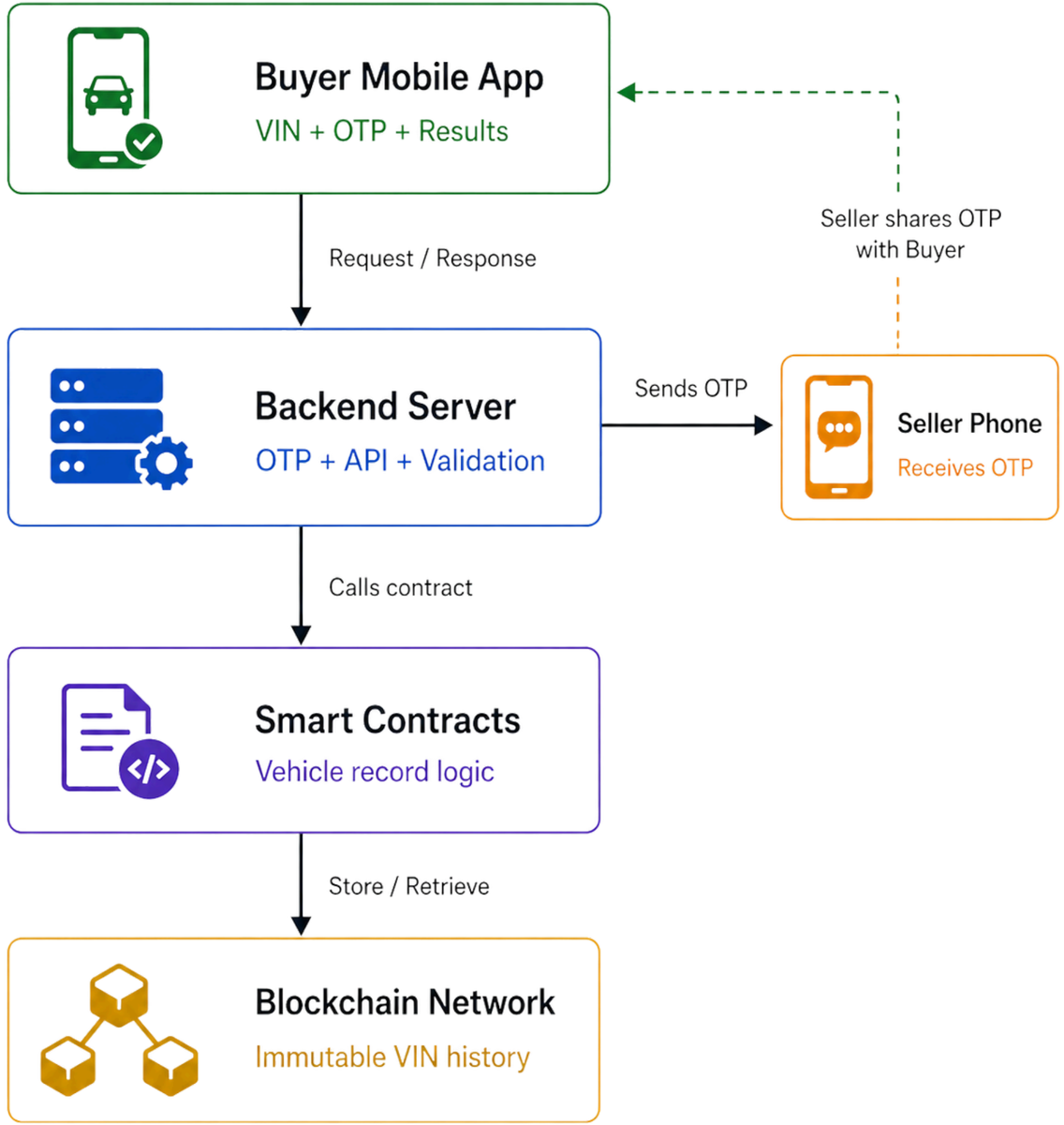
Data Flow

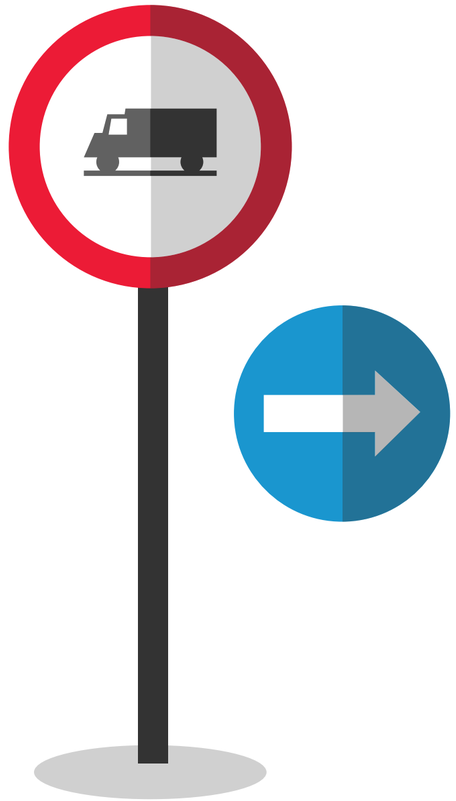

BUYER
1
Buyer App
(Enter VIN)
Request
(VIN Lookup)
2
Backend Server
OTP
3
SELLER
Seller Phone
(Receives OTP)
4
Buyer App
(Enter OTP)
OTP
(Shares OTP)
Verify
(OTP)
5
Backend Server
Query
6
Smart Contract
Retrieve Data
7
Blockchain Network
Response
8
Backend Server
Display Results
9
Buyer App
(Vehicle History Display)
Buyer Steps
Seller Steps
OTP Steps
Blockchain Steps
Backend Steps

User Flow


START
1
Buyer opens SijjilChain app
2
VIN / CHASSIS
1HGCM82633A004352
Buyer enters VIN / chassis number
3
OTP
SELLER
System sends OTP to seller
System sends OTP to seller
4
SELLER
123456
BUYER
Seller shares OTP with buyer
Seller shares OTP with buyer
5
123456
Buyer enters OTP
Buyer enters OTP
6
System verifies access
7
Buyer views vehicle report
8
Buyer checks accident history and risk status
END

System Architecture Diagram





08 User Experience (UX)

User Experience (UX)

User-Friendly Interface

Sijjil Chain has an easy-to-use, straightforward design allowing for fast access to vehicle information.

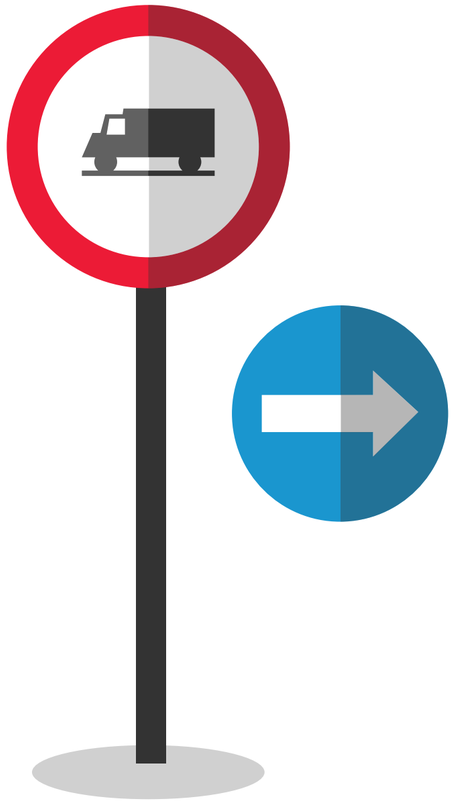
Step-by-step process

There are **three basic steps** to complete the task of **viewing the history** of your vehicle:

1. VIN Number (Vehicle Identification Number)
2. Send OTP (one-time password) to verify access;
3. Display Vehicle History

Clear layout

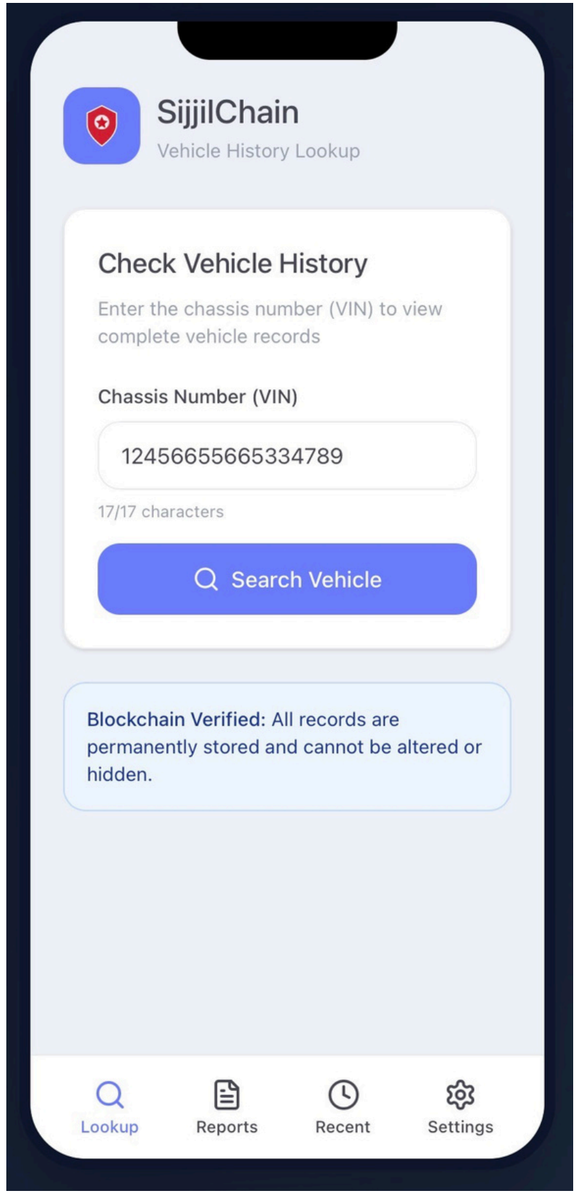
The use of clean, readable layouts and concise labeling along with icons and colors helps the user clearly see how well the vehicle has been maintained.

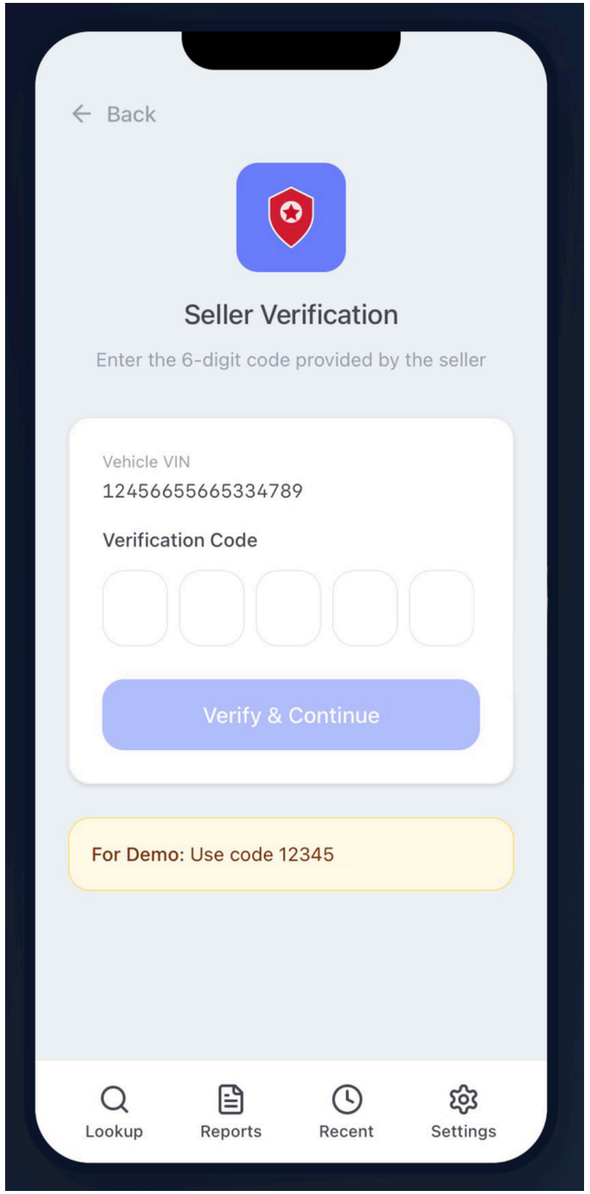


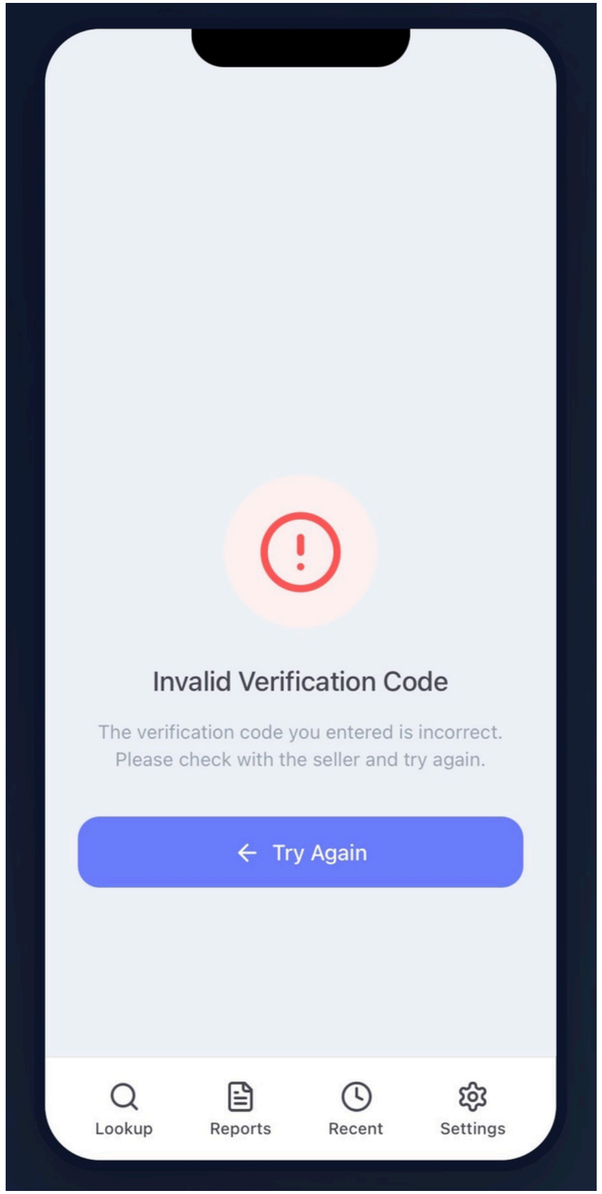
09 Example Screens

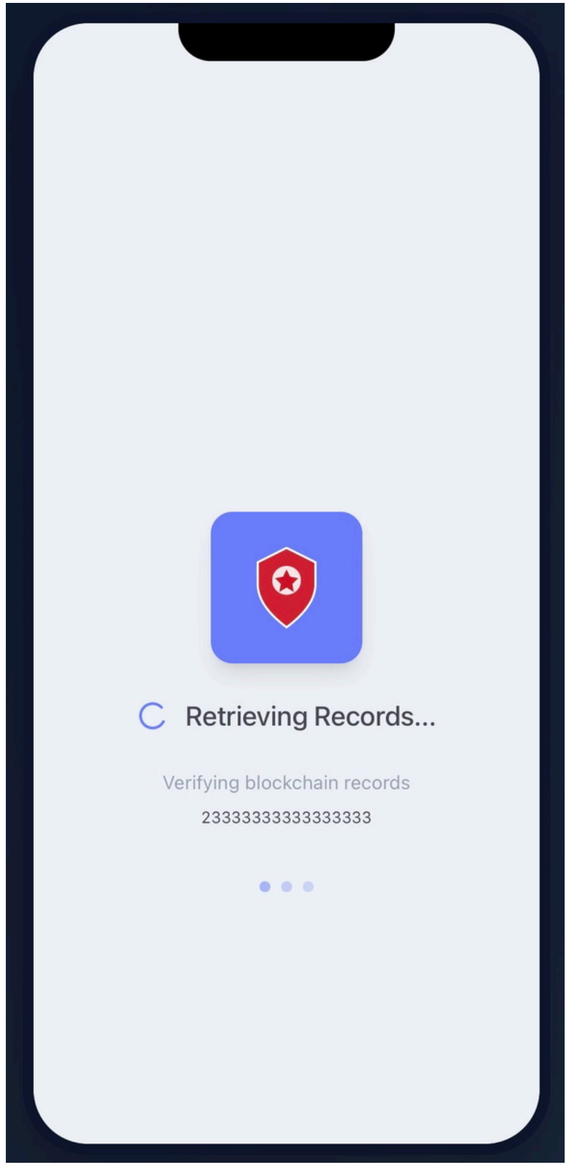
Inside this section:

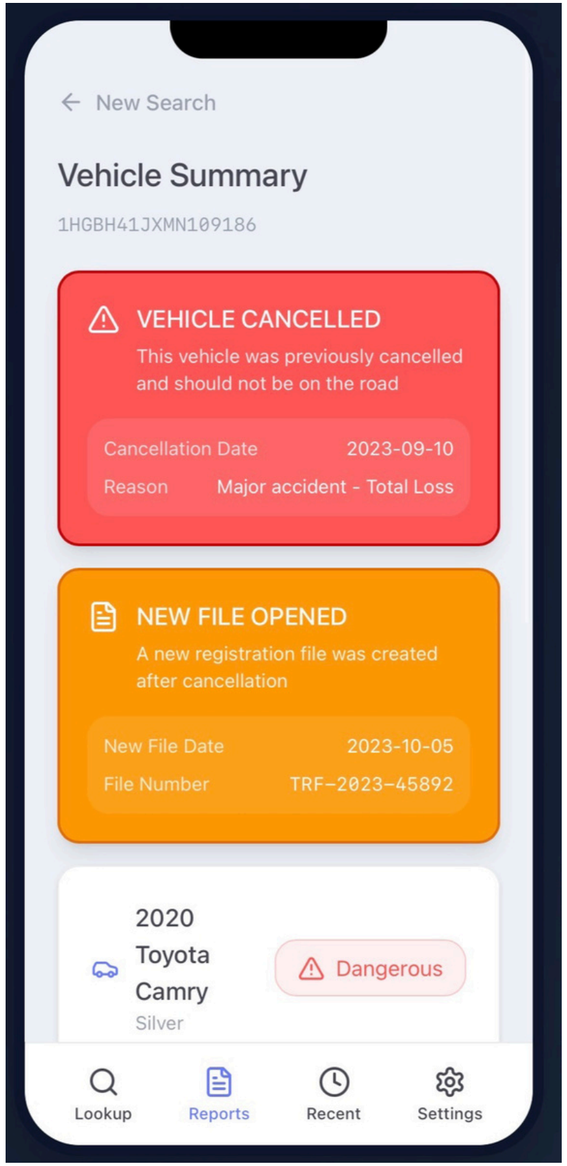
Snapshots of the application

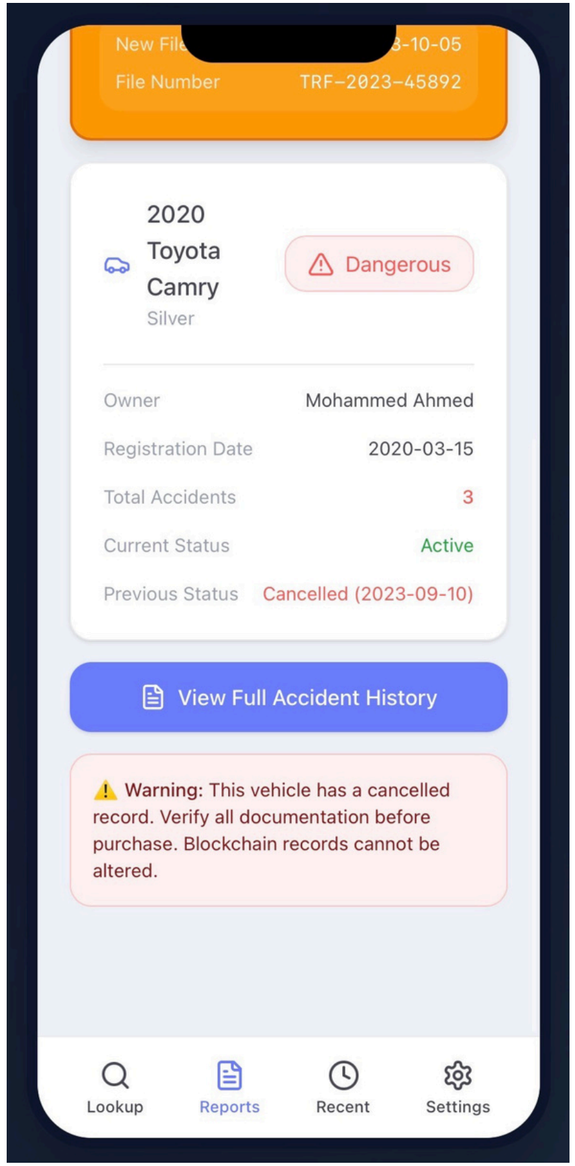

SijjilChain
Vehicle History Lookup
Check Vehicle History
Enter the chassis number (VIN) to view complete vehicle records
Chassis Number (VIN)
12456655665334789
17/17 characters
Search Vehicle
Blockchain Verified: All records are permanently stored and cannot be altered or hidden.
Lookup
Reports
Recent
Settings

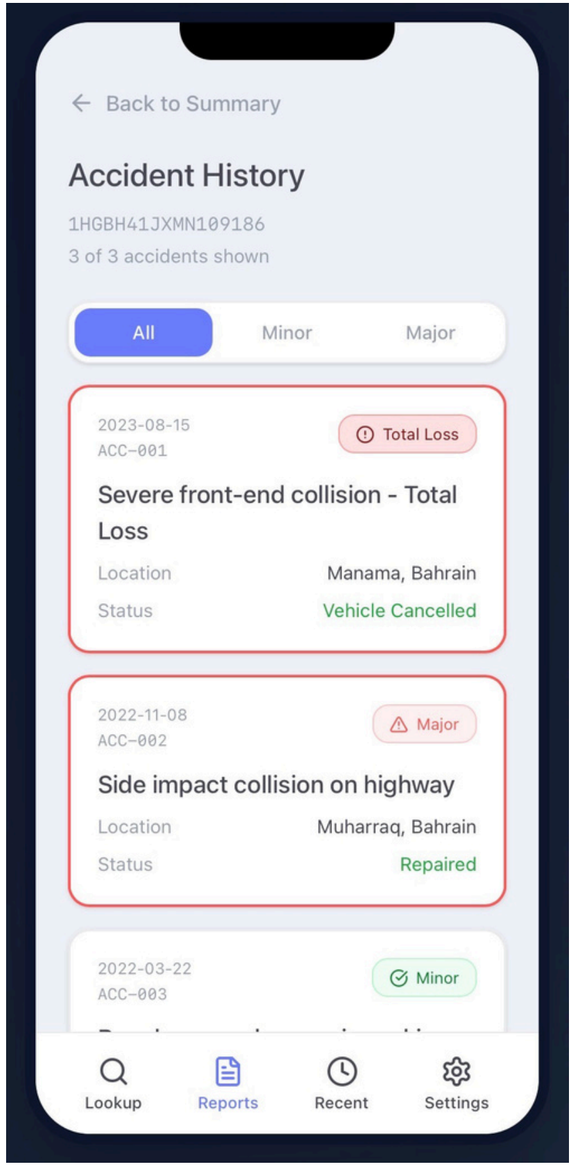

Back
Seller Verification
Enter the 6-digit code provided by the seller
Vehicle VIN
12456655665334789
Verification Code
Verify & Continue
For Demo: Use code 12345
Lookup
Reports
Recent
Settings

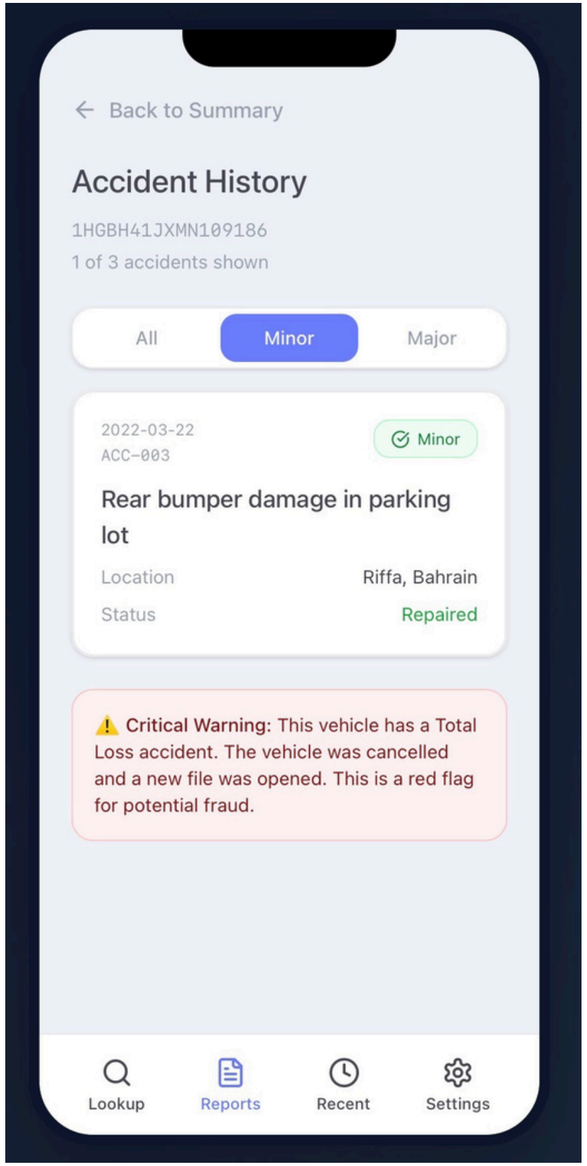

Invalid Verification Code
The verification code you entered is incorrect. Please check with the seller and try again.
Try Again
Lookup
Reports
Recent
Settings

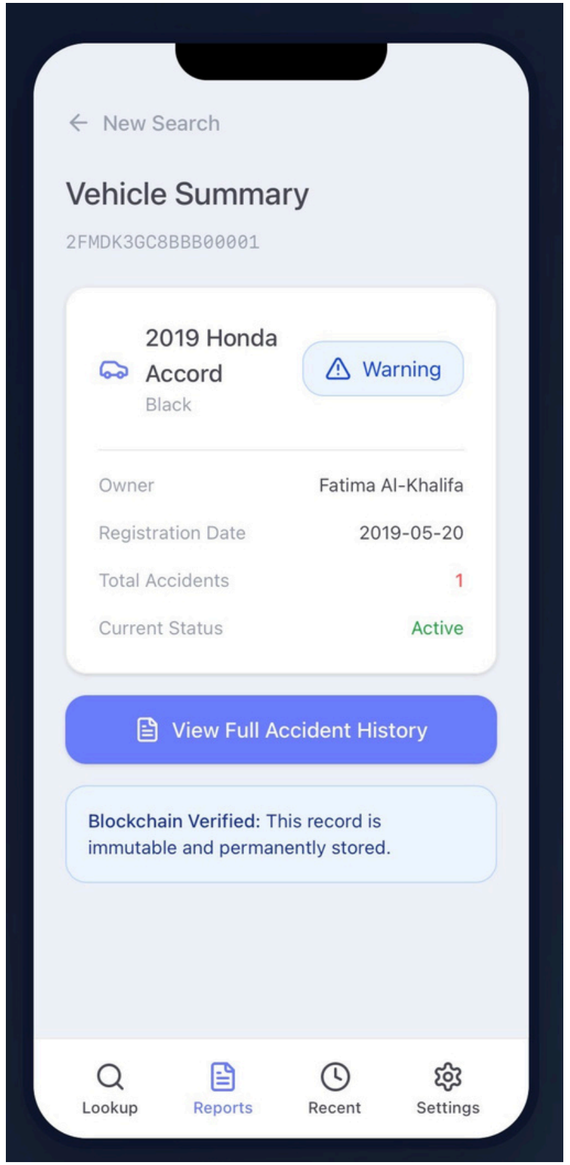

Retrieving Records...
Verifying blockchain records
23333333333333333

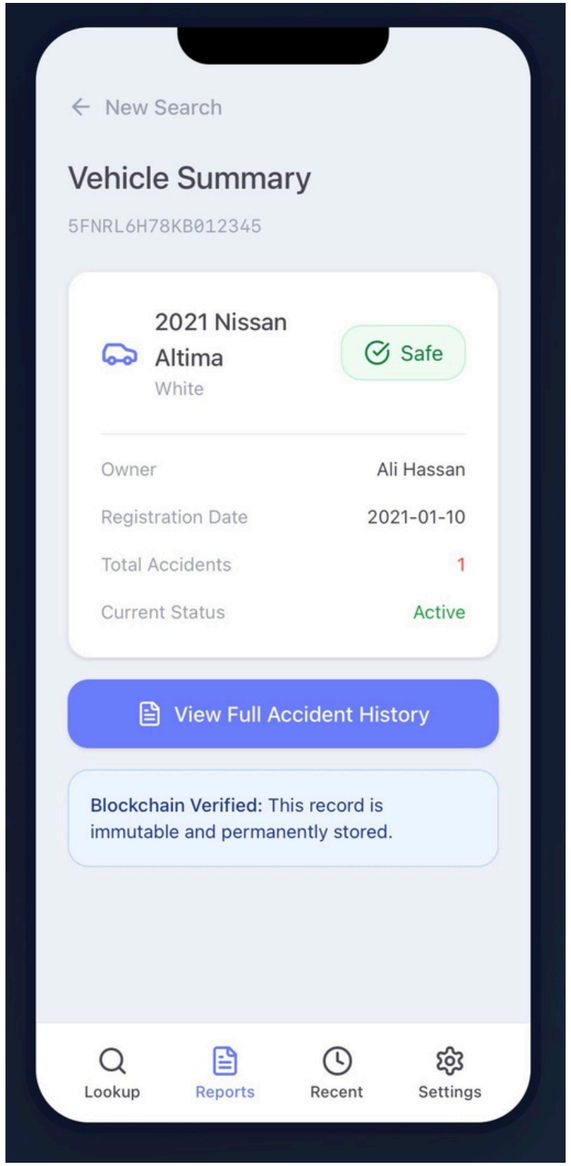

New Search
Vehicle Summary
1HGBH41JXMN109186
VEHICLE CANCELLED
This vehicle was previously cancelled and should not be on the road
Cancellation Date 2023-09-10
Reason Major accident - Total Loss
NEW FILE OPENED
A new registration file was created after cancellation
New File Date 2023-10-05
File Number TRF-2023-45892
2020 Toyota Camry
Silver
Dangerous
Lookup
Reports
Recent
Settings

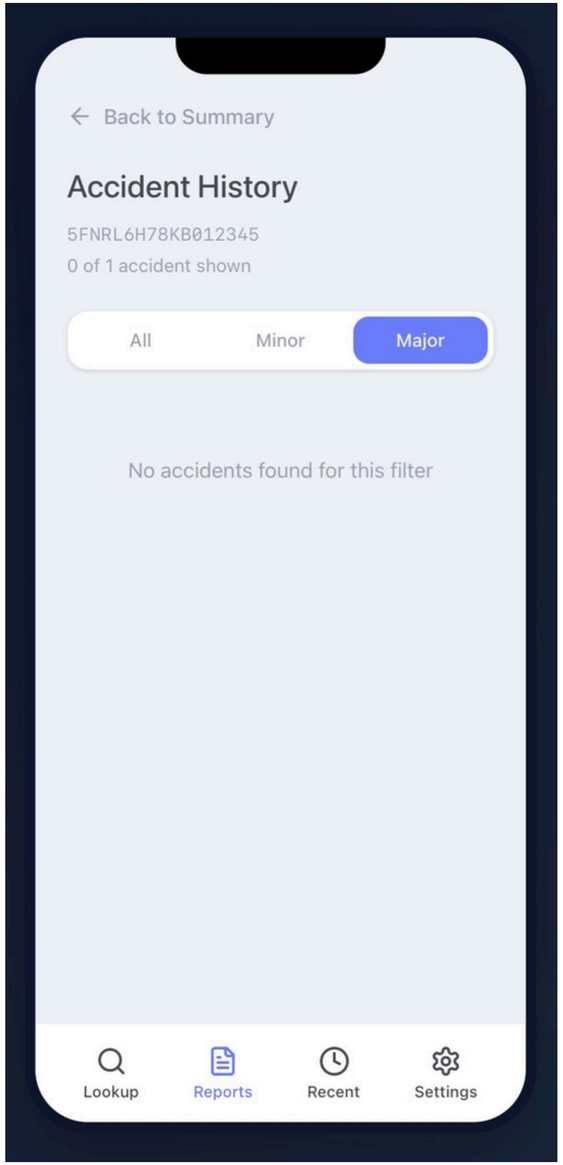

File Number TRF-2023-45892
2020 Toyota Camry
Silver
Dangerous
Owner Mohammed Ahmed
Registration Date 2020-03-15
Total Accidents 3
Current Status Active
Previous Status Cancelled (2023-09-10)
View Full Accident History
Warning: This vehicle has a cancelled record. Verify all documentation before purchase. Blockchain records cannot be altered.
Lookup
Reports
Recent
Settings

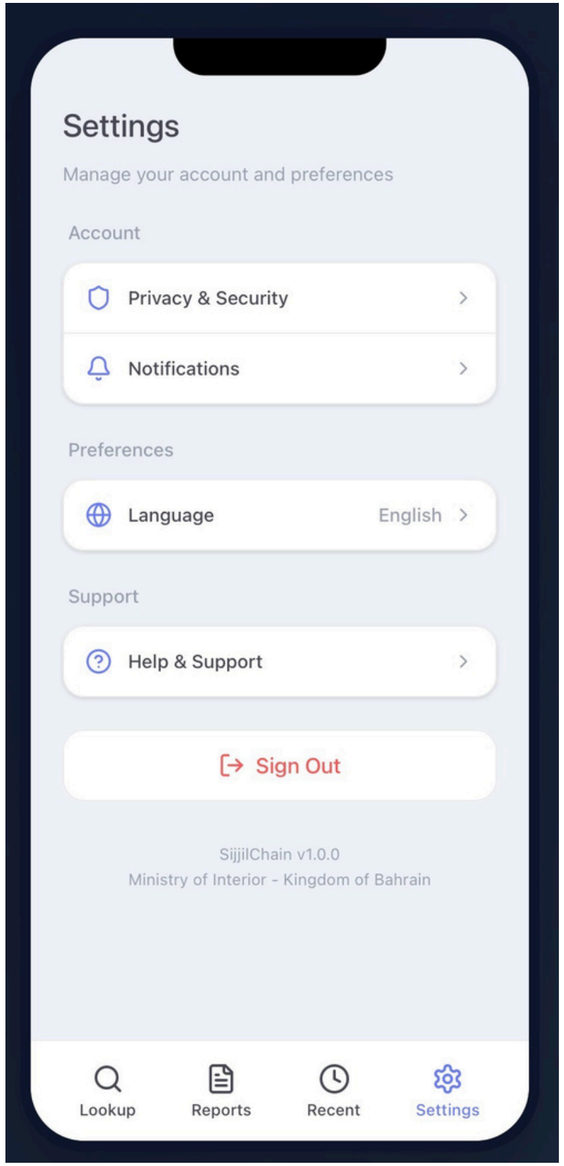

Back to Summary
Accident History
1HGBH41JXMN109186
3 of 3 accidents shown
All
Minor
Major
2023-08-15
ACC-001
Total Loss
Severe front-end collision - Total Loss
Location
Manama, Bahrain
Status
Vehicle Cancelled
2022-11-08
ACC-002
Major
Side impact collision on highway
Location
Muharraq, Bahrain
Status
Repaired
2022-03-22
ACC-003
Minor
Lookup
Reports
Recent
Settings

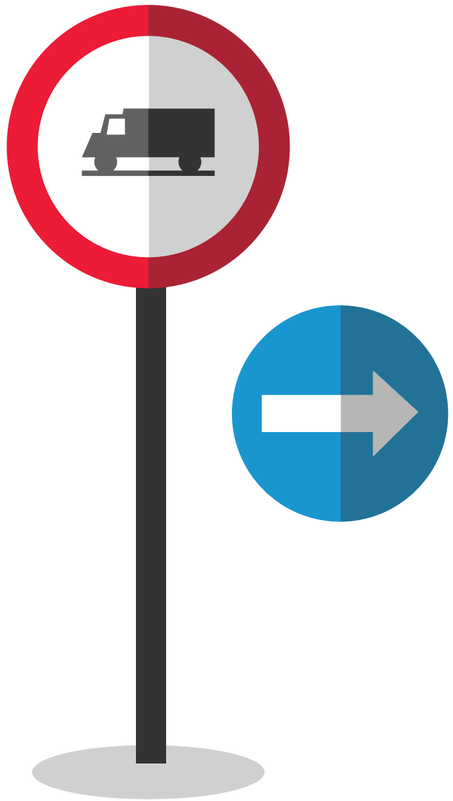

Back to Summary
Accident History
1HGBH41JXMN109186
1 of 3 accidents shown
All
Minor
Major
2022-03-22
ACC-003
Minor
Rear bumper damage in parking lot
Location
Riffa, Bahrain
Status
Repaired
⚠️ Critical Warning: This vehicle has a Total Loss accident. The vehicle was cancelled and a new file was opened. This is a red flag for potential fraud.
Lookup
Reports
Recent
Settings


New Search
Vehicle Summary
2FMDK3GC8BBB00001
2019 Honda Accord
Black
Warning
Owner
Fatima Al-Khalifa
Registration Date
2019-05-20
Total Accidents
1
Current Status
Active
View Full Accident History
Blockchain Verified: This record is immutable and permanently stored.
Lookup
Reports
Recent
Settings


New Search
Vehicle Summary
5FNRL6H78KB012345
2021 Nissan Altima
White
Safe
Owner
Ali Hassan
Registration Date
2021-01-10
Total Accidents
1
Current Status
Active
View Full Accident History
Blockchain Verified: This record is immutable and permanently stored.
Lookup
Reports
Recent
Settings


Back to Summary
Accident History
5FNRL6H78KB012345
0 of 1 accident shown
All
Minor
Major
No accidents found for this filter
Lookup
Reports
Recent
Settings


Settings
Manage your account and preferences
Account
Privacy & Security
Notifications
Preferences
Language
English
Support
Help & Support
Sign Out
SijjilChain v1.0.0
Ministry of Interior - Kingdom of Bahrain
Lookup
Reports
Recent
Settings

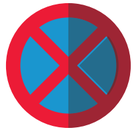


10 Impacts

Impacts

SijjilChain will have an enormous positive effect on the implementation of SijjilChain

Makes the used car industry more transparent



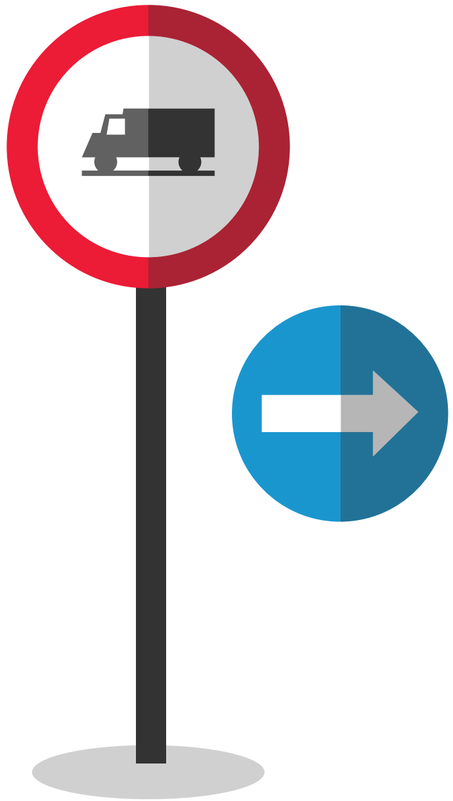
Stops all types of fraudulent activities with regards to used car's past history

Enables buyers to be able to make better choices when purchasing a vehicle

Saves buyers from having to physically go to traffic office to get information about vehicles

Will build confidence in government programs

Overall increases safety on the roads



11 Feasibility

Feasibility

Technical Feasibility

- Has to be able to utilize the available blockchain platforms
- Will integrate into the current governmental systems

Operational Feasibility

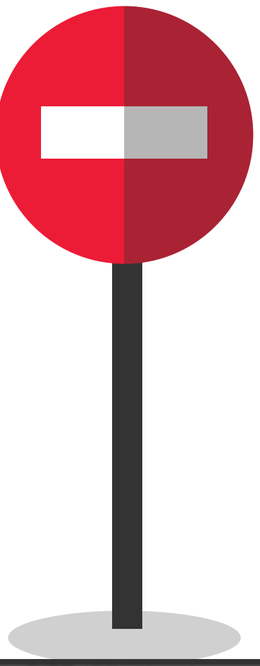
- Will integrate with the Traffic Directorate and Insurance Databases
- Can update Vehicle records in a real-time format.

Economic Feasibility

- Eliminates Manual Work and Administrative Costs.
- Enhances Service Delivery Efficiency.

Challenges

- Migration of data from the present system
- Regulatory and Legal Approval
- Adoption and Awareness by Users

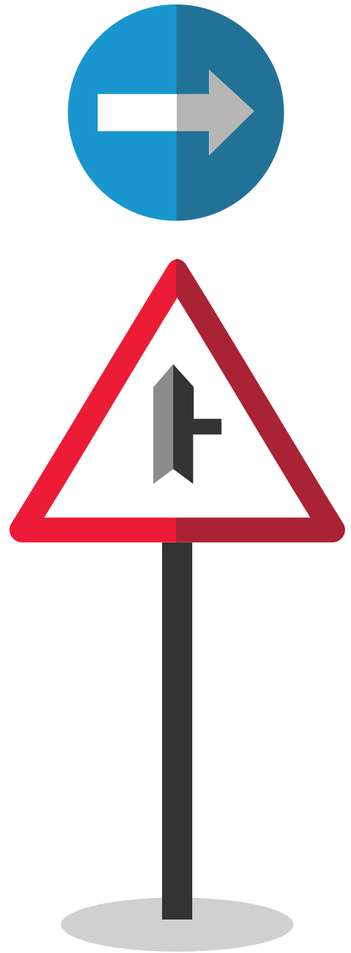


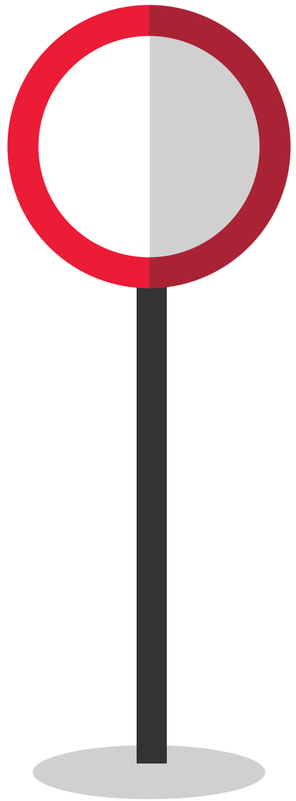
Conclusion

Sijjil Chain addresses the problem of vehicles **being sold** without their **history** being revealed; it does this through a **Blockchain based VIN** (Vehicle Identification Number) centered System that provides **Transparency & Trust**.

Through providing **one immutable record** for each Vehicle's history; **Fraud** will be prevented, therefore enabling Users to Make Informed Decisions.

The Solution is **Practical, Secure and Aligned** with the **Digital Transformation Goals** for Government Services in Bahrain.





Thank YOU